

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Polynomials

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the zeroes of $x^2 - 7x + 10$.

Answer:

$x^2 - 7x + 10 = (x - 2)(x - 5)$. The zeroes are 2 and 5.

Question 2

Find the zeroes of $x^2 - 1x - 12$.

Answer:

$x^2 - 1x - 12 = (x - (-3))(x - 4)$. The zeroes are -3 and 4.

Question 3

Find the zeroes of $x^2 + 5x - 6$.

Answer:

$x^2 + 5x - 6 = (x - 1)(x - (-6))$. The zeroes are 1 and -6.

Question 4

Find the zeroes of $x^2 - 10x + 21$.

Answer:

$x^2 - 10x + 21 = (x - 7)(x - 3)$. The zeroes are 7 and 3.

Question 5

Find the zeroes of $x^2 + 7x + 10$.

Answer:

$x^2 + 7x + 10 = (x - (-2))(x - (-5))$. The zeroes are -2 and -5.

Question 6

Find the zeroes of $x^2 - 10x + 24$.

Answer:

$x^2 - 10x + 24 = (x - 4)(x - 6)$. The zeroes are 4 and 6.

Question 7

Find the zeroes of $x^2 - 7x - 8$.

Answer:

$x^2 - 7x - 8 = (x - 8)(x - (-1))$. The zeroes are 8 and -1.

Question 8

Find the zeroes of $x^2 - 11x + 18$.

Answer:

$x^2 - 11x + 18 = (x - 9)(x - 2)$. The zeroes are 9 and 2.

Question 9

Form a quadratic polynomial with zeroes 2 and 3 and leading coefficient 2.

Answer:

$p(x) = 2[x^2 - (5)x + (6)] = 2x^2 - 10x + 12$.

Polynomials - Continued

Question 10

Form a quadratic polynomial with zeroes -1 and 4 and leading coefficient 3.

Answer:

$$p(x) = 3[x^2 - (3)x + (-4)] = 3x^2 - 9x - 12.$$

Question 11

Form a quadratic polynomial with zeroes 5 and -2 and leading coefficient 1.

Answer:

$$p(x) = 1[x^2 - (3)x + (-10)] = x^2 - 3x - 10.$$

Question 12

Form a quadratic polynomial with zeroes 6 and 1 and leading coefficient 4.

Answer:

$$p(x) = 4[x^2 - (7)x + (6)] = 4x^2 - 28x + 24.$$

Question 13

Form a quadratic polynomial with zeroes -3 and -4 and leading coefficient 2.

Answer:

$$p(x) = 2[x^2 - (-7)x + (12)] = 2x^2 - 14x + 24.$$

Question 14

Form a quadratic polynomial with zeroes 7 and -1 and leading coefficient 5.

Answer:

$$p(x) = 5[x^2 - (6)x + (-7)] = 5x^2 - 30x - 35.$$

Question 15

Find $p(2)$ for $p(x) = 2x^2 - 3x - 5$.

Answer:

$$p(2) = 2(2)^2 - 3(2) - 5 = -3.$$

Question 16

Find $p(2)$ for $p(x) = 1x^2 + 4x - 12$.

Answer:

$$p(2) = 1(2)^2 + 4(2) - 12 = 0.$$

Question 17

Find $p(-1)$ for $p(x) = 3x^2 - 5x - 2$.

Answer:

$$p(-1) = 3(-1)^2 - 5(-1) - 2 = 6.$$

Question 18

Find $p(-2)$ for $p(x) = 2x^2 + 7x + 3$.

Answer:

$$p(-2) = 2(-2)^2 + 7(-2) + 3 = -3.$$

Question 19

Find $p(3)$ for $p(x) = 4x^2 - 4x - 8$.

Answer:

$$p(3) = 4(3)^2 - 4(3) - 8 = 16.$$

Polynomials - Continued

Question 20

Find $p(2)$ for $p(x) = 5x^2 + 0x - 20$.

Answer:

$$p(2) = 5(2)^2 + 0(2) - 20 = 0.$$

Question 21

State the main formula, rule or theorem used in Polynomials.

Answer:

For $ax^2 + bx + c$, sum of zeroes = $-b/a$ and product = c/a .

Question 22

What do the important symbols or quantities in Polynomials represent?

Answer:

A zero is an x-value for which $p(x)=0$; it is also an x-intercept of the graph.

Question 23

Write the correct method for solving a Polynomials numerical.

Answer:

Factorise, use the quadratic formula where needed, then verify sum and product.

Question 24

Which unit, sign convention or condition must be checked in a Polynomials answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Polynomials be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Polynomials questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Polynomials numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Polynomials?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Polynomials - Continued

Question 29

How can a student decide which formula to use in Polynomials?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Polynomials?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

ONE SHOT MASTERY